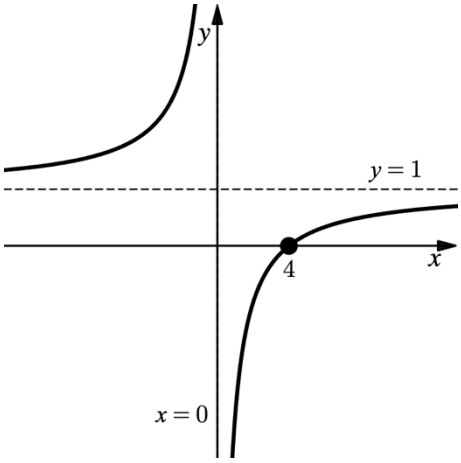
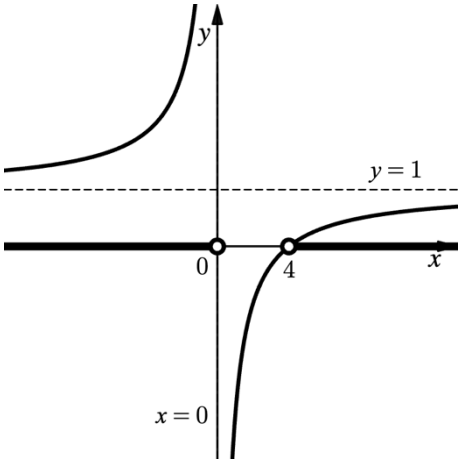
1) a)

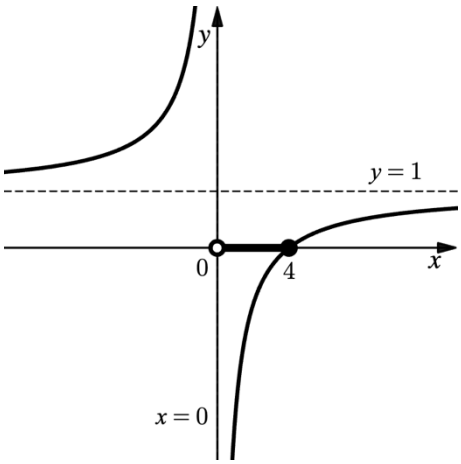


1) b) i)



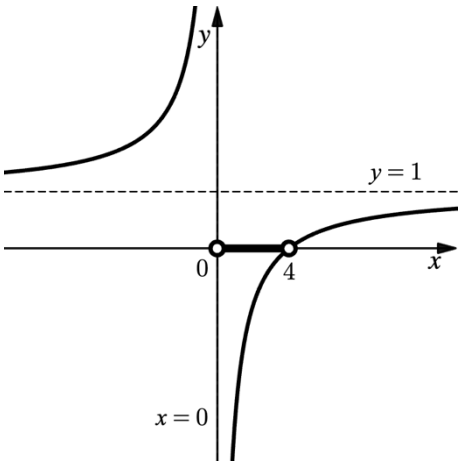
$\{x: x \in \mathbb{R}, x < 0\} \cup \{x: x \in \mathbb{R}, x > 4\}$

1) b) ii)



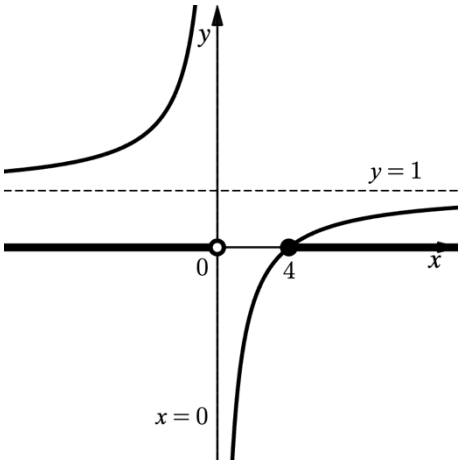
$\{x: x \in \mathbb{R}, 0 < x \leq 4\}$

1) c) i)



$x \in (0, 4)$

1) c) ii)



$x \in (-\infty, 0) \cup [4, \infty)$

2) a)

$\frac{1}{x} > 0$

Only critical value is $x = 0$

$x < 0$	$x > 0$
$-$	$+$

$\{x: x \in \mathbb{R}, x > 0\}$

2) b)

$\frac{1}{x-3} < 0$

Only critical value is $x = 3$

$x < 3$	$x > 3$
$-$	$+$

$\{x: x \in \mathbb{R}, x < 3\}$

2) c)

$\frac{2}{x+1} \geq 0$

Only critical value is $x = -1$

$x < -1$	$x > -1$
$-$	$+$

$\{x: x \in \mathbb{R}, x > -1\}$

2) d)

$\frac{5}{2x+3} \leq 0$

Only critical value is $x = -\frac{3}{2}$

$x < -\frac{3}{2}$	$x > -\frac{3}{2}$
$-$	$+$

$\left\{x: x \in \mathbb{R}, x < -\frac{3}{2}\right\}$

2) e)

$\frac{8}{4-7x} > 0$

Only critical value is $x = \frac{4}{7}$

$x < \frac{4}{7}$	$x > \frac{4}{7}$
$+$	$-$

$\left\{x: x \in \mathbb{R}, x < \frac{4}{7}\right\}$

3) a)

$\frac{1}{x} > 2 \Rightarrow \frac{1}{x} \times x^2 > 2 \times x^2 \Rightarrow x > 2x^2 \Rightarrow x - 2x^2 > 0$

$\Rightarrow (1 - 2x)x > 0$

Critical values at $x = 0$ and $x = \frac{1}{2}$

$x < 0$	$0 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$-$	$+$	$-$

$x \in \left(0, \frac{1}{2}\right)$

3) b)

$\frac{1}{x-2} < 3$

$\Rightarrow \frac{1}{x-2} \times (x-2)^2 < 3 \times (x-2)^2$

$\Rightarrow x - 2 < 3(x-2)^2$

$\Rightarrow (x-2) - 3(x-2)^2 < 0$

$\Rightarrow \big(1 - 3(x-2)\big)(x-2) < 0$

$\Rightarrow (7 - 3x)(x-2) < 0$

Critical values at $x = 2$ and $x = \frac{7}{3}$

$x < 2$	$2 < x < \frac{7}{3}$	$x > \frac{7}{3}$
$-$	$+$	$-$

$x \in (-\infty, 2) \cup \left(\frac{7}{3}, \infty\right)$

3) c)

$\frac{3}{x+4} \geq -5$

$\Rightarrow \frac{3}{x+4} \times (x+4)^2 \geq -5 \times (x+4)^2$

$\Rightarrow 3(x+4) \geq -5(x+4)^2$

$\Rightarrow 3(x+4) + 5(x+4)^2 \geq 0$

$\Rightarrow \big(3 + 5(x+4)\big)(x+4) \geq 0$

$\Rightarrow (5x+23)(x+4) \geq 0$

Critical values at $x = -\frac{23}{5}$ and $x = -4$

$x < -\frac{23}{5}$	$-\frac{23}{5} < x < -4$	$x > -4$
$+$	$-$	$+$

$x \in \left(-\infty, -\frac{23}{5}\right] \cup (-4, \infty)$

3) d)

$\frac{4}{3-2x} \leq 7$

$\Rightarrow \frac{4}{3-2x} \times (3-2x)^2 \leq 7 \times (3-2x)^2$

$\Rightarrow 4(3-2x) \leq 7(3-2x)^2$

$\Rightarrow 4(3-2x) - 7(3-2x)^2 \leq 0$

$\Rightarrow \big(4 - 7(3-2x)\big)(3-2x) \leq 0$

$\Rightarrow (14x-17)(3-2x) \leq 0$

Critical values at $x = \frac{17}{14}$ and $x = \frac{3}{2}$

$x < \frac{17}{14}$	$\frac{17}{14} < x < \frac{3}{2}$	$x > \frac{3}{2}$
$-$	$+$	$-$

$x \in \left(-\infty, \frac{17}{14}\right] \cup \left(\frac{3}{2}, \infty\right)$

3) e)

$\frac{10}{6-3x} > 9$

$\Rightarrow \frac{10}{6-3x} \times (6-3x)^2 > 9 \times (6-3x)^2$

$\Rightarrow 10(6-3x) > 9(6-3x)^2$

$\Rightarrow 10(6-3x) - 9(6-3x)^2 > 0$

$\Rightarrow \big(10 - 9(6-3x)\big)(6-3x) > 0$

$\Rightarrow (27x-44)(6-3x) > 0$

$\Rightarrow (27x-44)(2-x) > 0$

Critical values at $x = \frac{44}{27}$ and $x = 2$

$x < \frac{44}{27}$	$\frac{44}{27} < x < 2$	$x > 2$
$-$	$+$	$-$

$x \in \left(\frac{44}{27}, 2\right)$

4) a)

The error was made when Francesca multiplied both sides of the inequality by $(x-3)$. As $(x-3)$ could be negative, this would change the direction of the inequality. Francesca should instead multiply both sides of the inequality by $(x-3)^2$ at this stage.

4) b)

$\frac{5}{x-3} \geq 9 - x$

$\Rightarrow \frac{5}{x-3} \times (x-3)^2 \geq (9-x) \times (x-3)^2$

$\Rightarrow 5(x-3) \geq (9-x)(x-3)^2$

$\Rightarrow 5(x-3) - (9-x)(x-3)^2 \geq 0$

$\Rightarrow \big(5 - (9-x)(x-3)\big)(x-3) \geq 0$

$\Rightarrow \big(5 - (-x^2 + 12x - 27)\big)(x-3) \geq 0$

$\Rightarrow (x^2 - 12x + 32)(x-3) \geq 0$

$\Rightarrow (x-4)(x-8)(x-3) \geq 0$

Critical values at $x = 3$, $x = 4$ and $x = 8$

$x < 3$	$3 < x < 4$	$4 < x < 8$	$x > 8$
$-$	$+$	$-$	$+$

$x \in (3,4] \cup [8, \infty)$

5) a)

$\frac{x+1}{x} > 0$

Critical values at $x = -1$ and $x = 0$

$x < -1$	$-1 < x < 0$	$x > 0$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x < -1\} \cup \{x: x \in \mathbb{R}, x > 0\}$

5) b)

$\frac{x-1}{2x} < 0$

Critical values at $x = 0$ and $x = 1$

$x < 0$	$0 < x < 1$	$x > 1$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, 0 < x < 1\}$

5) c)

$\frac{2x-3}{4x} \geq 0$

Critical values at $x = 0$ and $x = \frac{3}{2}$

$x < 0$	$0 < x < \frac{3}{2}$	$x > \frac{3}{2}$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x < 0\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{3}{2}\right\}$

5) d)

$\frac{5-3x}{6x} \leq 0$

Critical values at $x = 0$ and $x = \frac{5}{3}$

$x < 0$	$0 < x < \frac{5}{3}$	$x > \frac{5}{3}$
$-$	$+$	$-$

$\{x: x \in \mathbb{R}, x < 0\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{5}{3}\right\}$

6) a)

$\frac{x+2}{x} > 1$

$\Rightarrow \frac{x+2}{x} \times x^2 > 1 \times x^2$

$\Rightarrow (x+2)x > x^2$

$\Rightarrow (x+2)x - x^2 > 0$

$\Rightarrow \big((x+2) - x\big)x > 0$

$\Rightarrow 2x > 0$

$\Rightarrow x > 0$

$x \in (0, \infty)$

6) b)

$\frac{x-4}{x} < 1$

$\Rightarrow \frac{x-4}{x} \times x^2 < 1 \times x^2$

$\Rightarrow (x-4)x < x^2$

$\Rightarrow (x-4)x - x^2 < 0$

$\Rightarrow \big((x-4) - x\big)x < 0$

$\Rightarrow -4x < 0$

$\Rightarrow x > 0$

$x \in (0, \infty)$

6) c)

$\frac{2x+1}{2x} \geq 2$

$\Rightarrow \frac{2x+1}{2x} \times x^2 \geq 2 \times x^2$

$\Rightarrow (2x+1)x \geq 4x^2$

$\Rightarrow (2x+1)x - 4x^2 \geq 0$

$\Rightarrow \big((2x+1) - 4x\big)x \geq 0$

$\Rightarrow (1-2x)x \geq 0$

Critical values at $x = 0$ and $x = \frac{1}{2}$

$x < 0$	$0 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$-$	$+$	$-$

$x \in \left(0, \frac{1}{2}\right]$

6) d)

$\frac{5-3x}{x} \leq -4$

$\Rightarrow \frac{5-3x}{x} \times x^2 \leq -4 \times x^2$

$\Rightarrow (5-3x)x \leq -4x^2$

$\Rightarrow (5-3x)x + 4x^2 \leq 0$

$\Rightarrow \big((5-3x)+4x\big)x \leq 0$

$\Rightarrow (5+x)x \leq 0$

Critical values at $x = -5$ and $x = 0$

$x < -5$	$-5 < x < 0$	$x > 0$
$+$	$-$	$+$

$x \in [-5, 0)$

7) a)

$\frac{x}{x+1} > 0$

Critical values at $x = -1$ and $x = 0$

$x < -1$	$-1 < x < 0$	$x > 0$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x < -1\} \cup \{x: x \in \mathbb{R}, x > 0\}$

7) b)

$\frac{2x}{x-3} < 0$

Critical values at $x = 0$ and $x = 3$

$x < 0$	$0 < x < 3$	$x > 3$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, 0 < x < 3\}$

7) c)

$\frac{4x}{2x+3} \geq 0$

Critical values at $x = -\frac{3}{2}$ and $x = 0$

$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 0$	$x > 0$
$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, x < -\frac{3}{2}\right\} \cup \{x: x \in \mathbb{R}, x \geq 0\}$

7) d)

$\frac{3x}{3-x} \leq 0$

Critical values at $x = 0$ and $x = 3$

$x < 0$	$0 < x < 3$	$x > 3$
$-$	$+$	$-$

$\{x: x \in \mathbb{R}, x \leq 0\} \cup \{x: x \in \mathbb{R}, x > 3\}$

8) a)

$\frac{x-3}{x+1} > 0$

Critical values at $x = -1$ and $x = 3$

$x < -1$	$-1 < x < 3$	$x > 3$
$+$	$-$	$+$

$x \in (-\infty, -1) \cup (3, \infty)$

8) b)

$\frac{x+4}{x-5} < 0$

Critical values at $x = -4$ and $x = 5$

$x < -4$	$-4 < x < 5$	$x > 5$
$+$	$-$	$+$

$x \in (-4, 5)$

8) c)

$\frac{2x+1}{x+3} \geq 0$

Critical values at $x = -3$ and $x = -\frac{1}{2}$

$x < -3$	$-3 < x < -\frac{1}{2}$	$x > -\frac{1}{2}$
$+$	$-$	$+$

$x \in (-\infty, -3) \cup \left[-\frac{1}{2}, \infty\right)$

8) d)

$\frac{5-x}{2x+5} \leq 0$

Critical values at $x = -\frac{5}{2}$ and $x = 5$

$x < -\frac{5}{2}$	$-\frac{5}{2} < x < 5$	$x > 5$
$-$	$+$	$-$

$x \in \left(-\infty, -\frac{5}{2}\right) \cup [5, \infty)$

9) a)

$\frac{x+3}{x-2} > 1$

$\Rightarrow \frac{x+3}{x-2} \times (x-2)^2 > 1 \times (x-2)^2$

$\Rightarrow (x+3)(x-2) > (x-2)^2$

$\Rightarrow (x+3)(x-2) - (x-2)^2 > 0$

$\Rightarrow \big((x+3) - (x-2)\big)(x-2) > 0$

$\Rightarrow 5(x-2) > 0$

$\Rightarrow x - 2 > 0$

$\Rightarrow x > 2$

$\{x: x \in \mathbb{R}, x > 2\}$

9) b)

$\frac{x-6}{x+1} < 1$

$\Rightarrow \frac{x-6}{x+1} \times (x+1)^2 < 1 \times (x+1)^2$

$\Rightarrow (x-6)(x+1) < (x+1)^2$

$\Rightarrow (x-6)(x+1) - (x+1)^2 < 0$

$\Rightarrow \big((x-6) - (x+1)\big)(x+1) < 0$

$\Rightarrow -7(x+1) < 0$

$\Rightarrow x + 1 > 0$

$\Rightarrow x > -1$

$\{x: x \in \mathbb{R}, x > -1\}$

9) c)

$\frac{x+4}{2x-1} \geq -3$

$\Rightarrow \frac{x+4}{2x-1} \times (2x-1)^2 \geq -3 \times (2x-1)^2$

$\Rightarrow (x+4)(2x-1) \geq -3(2x-1)^2$

$\Rightarrow (x+4)(2x-1) + 3(2x-1)^2 \geq 0$

$\Rightarrow \big((x+4) + 3(2x-1)\big)(2x-1) \geq 0$

$\Rightarrow (7x+1)(2x-1) \geq 0$

Critical values at $x = -\frac{1}{7}$ and $x = \frac{1}{2}$

$x < -\frac{1}{7}$	$-\frac{1}{7} < x < \frac{1}{2}$	$x > \frac{1}{2}$
$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, x \leq -\frac{1}{7}\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{1}{2}\right\}$

9) d)

$\frac{7-x}{x-8} \leq 5$

$\Rightarrow \frac{7-x}{x-8} \times (x-8)^2 \leq 5 \times (x-8)^2$

$\Rightarrow (7-x)(x-8) \leq 5(x-8)^2$

$\Rightarrow (7-x)(x-8) - 5(x-8)^2 \leq 0$

$\Rightarrow \big((7-x) - 5(x-8)\big)(x-8) \leq 0$

$\Rightarrow (47-6x)(x-8) \leq 0$

Critical values at $x = \frac{47}{6}$ and $x = 8$

$x < \frac{47}{6}$	$\frac{47}{6} < x < 8$	$x > 8$
$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, x \leq \frac{47}{6}\right\} \cup \{x: x \in \mathbb{R}, x > 8\}$

10) a)

First solve $\frac{2x+5}{x-5} > 0$

Critical values at $x = -\frac{5}{2}$ and $x = 5$

$x < -\frac{5}{2}$	$-\frac{5}{2} < x < 5$	$x > 5$
$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, x < -\frac{5}{2}\right\} \cup \{x: x \in \mathbb{R}, x > 5\}$

Now solve $\frac{2x+5}{x-5} < 1$

$\Rightarrow \frac{2x+5}{x-5} \times (x-5)^2 < 1 \times (x-5)^2$

$\Rightarrow (2x+5)(x-5) < (x-5)^2$

$\Rightarrow (2x+5)(x-5) - (x-5)^2 < 0$

$\Rightarrow \big((2x+5) - (x-5)\big)(x-5) < 0$

$\Rightarrow (x+10)(x-5) < 0$

Critical values at $x = -10$ and $x = 5$

$x < -10$	$-10 < x < 5$	$x > 5$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, -10 < x < 5\}$

Both inequalities are true when $\left\{x: x \in \mathbb{R}, -10 < x < -\frac{5}{2}\right\}$

10) b)

First solve $\frac{3x+1}{2-x} > -1$

$\Rightarrow \frac{3x+1}{2-x} \times (2-x)^2 > -1 \times (2-x)^2$

$\Rightarrow (3x+1)(2-x) > -(2-x)^2$

$\Rightarrow (3x+1)(2-x) + (2-x)^2 > 0$

$\Rightarrow \big((3x+1) + (2-x)\big)(2-x) > 0$

$\Rightarrow (2x+3)(2-x) > 0$

Critical values at $x = -\frac{3}{2}$ and $x = 2$

$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 2$	$x > 2$
$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, -\frac{3}{2} < x < 2\right\}$

Now solve $\frac{3x+1}{2-x} < 1$

$\Rightarrow \frac{3x+1}{2-x} \times (2-x)^2 < 1 \times (2-x)^2$

$\Rightarrow (3x+1)(2-x) < (2-x)^2$

$\Rightarrow (3x+1)(2-x) - (2-x)^2 < 0$

$\Rightarrow \big((3x+1) - (2-x)\big)(2-x) < 0$

$\Rightarrow (4x-1)(2-x) < 0$

Critical values at $x = \frac{1}{4}$ and $x = 2$

$x < \frac{1}{4}$	$\frac{1}{4} < x < 2$	$x > 2$
$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, x < \frac{1}{4}\right\} \cup \{x: x \in \mathbb{R}, x > 2\}$

Both inequalities are true when $\left\{x: x \in \mathbb{R}, -\frac{3}{2} < x < \frac{1}{4}\right\}$

10) c)

First solve $\frac{5-x}{4x+1} \geq -2$

$\Rightarrow \frac{5-x}{4x+1} \times (4x+1)^2 \geq -2 \times (4x+1)^2$

$\Rightarrow (5-x)(4x+1) \geq -2(4x+1)^2$

$\Rightarrow (5-x)(4x+1) + 2(4x+1)^2 \geq 0$

$\Rightarrow \big((5-x) + 2(4x+1)\big)(4x+1) \geq 0$

$\Rightarrow (7x+7)(4x+1) \geq 0$

$\Rightarrow (x+1)(4x+1) \geq 0$

Critical values at $x = -1$ and $x = -\frac{1}{4}$

$x < -1$	$-1 < x < -\frac{1}{4}$	$x > -\frac{1}{4}$
$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x \leq -1\} \cup \left\{x: x \in \mathbb{R}, x \geq -\frac{1}{4}\right\}$

Now solve $\frac{5-x}{4x+1} \leq 2$

$\Rightarrow \frac{5-x}{4x+1} \times (4x+1)^2 \leq 2 \times (4x+1)^2$

$\Rightarrow (5-x)(4x+1) \leq 2(4x+1)^2$

$\Rightarrow (5-x)(4x+1) - 2(4x+1)^2 \leq 0$

$\Rightarrow \big((5-x) - 2(4x+1)\big)(4x+1) \leq 0$

$\Rightarrow (3-9x)(4x+1) \leq 0$

$\Rightarrow (1-3x)(4x+1) \leq 0$

Critical values at $x = -\frac{1}{4}$ and $x = \frac{1}{3}$

$x < -\frac{1}{4}$	$-\frac{1}{4} < x < \frac{1}{3}$	$x > \frac{1}{3}$
$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, x < -\frac{1}{4}\right\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{1}{3}\right\}$

Both inequalities are true when $\{x: x \in \mathbb{R}, x \leq -1\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{1}{3}\right\}$

11) a)

$\frac{2}{x} < 2x - 3$

$\Rightarrow \frac{2}{x} \times x^2 < (2x-3) \times x^2$

$\Rightarrow 2x < (2x-3)x^2$

$\Rightarrow 2x - (2x-3)x^2 < 0$

$\Rightarrow (2 - (2x-3)x)x < 0$

$\Rightarrow (2 + 3x - 2x^2)x < 0$

$\Rightarrow (2x+1)(2-x)x < 0$

Critical values at $x = -\frac{1}{2}$, $x = 0$ and $x = 2$

$x < -\frac{1}{2}$	$-\frac{1}{2} < x < 0$	$0 < x < 2$	$x > 2$
$+$	$-$	$+$	$-$

$x \in \left(-\frac{1}{2}, 0\right) \cup (2, \infty)$

11) b)

$\frac{5}{2x+1} \leq 3x - 2$

$\Rightarrow \frac{5}{2x+1} \times (2x+1)^2 \leq (3x-2) \times (2x+1)^2$

$\Rightarrow 5(2x+1) \leq (3x-2)(2x+1)^2$

$\Rightarrow 5(2x+1) - (3x-2)(2x+1)^2 \leq 0$

$\Rightarrow \big(5 - (3x-2)(2x+1)\big)(2x+1) \leq 0$

$\Rightarrow (-6x^2 + x + 7)(2x+1) \leq 0$

$\Rightarrow (7-6x)(x+1)(2x+1) \leq 0$

Critical values at $x = -1$, $x = -\frac{1}{2}$ and $x = \frac{7}{6}$

$x < -1$	$-1 < x < -\frac{1}{2}$	$-\frac{1}{2} < x < \frac{7}{6}$	$x > \frac{7}{6}$
$+$	$-$	$+$	$-$

$x \in \left[-1, -\frac{1}{2}\right) \cup \left[\frac{7}{6}, \infty\right)$

11) c)

$\frac{x+3}{x} > x - 1$

$\Rightarrow \frac{x+3}{x} \times x^2 > (x-1) \times x^2$

$\Rightarrow (x+3)x > (x-1)x^2$

$\Rightarrow (x+3)x - (x-1)x^2 > 0$

$\Rightarrow \big((x+3) - (x-1)x\big)x > 0$

$\Rightarrow (3 + 2x - x^2)x > 0$

$\Rightarrow (x+1)(3-x)x > 0$

Critical values at $x = -1$, $x = 0$ and $x = 3$

$x < -1$	$-1 < x < 0$	$0 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$

$x \in (-\infty, -1) \cup (0, 3)$

11) d)

$\frac{x-5}{2x} < 2 - x$

$\Rightarrow \frac{x-5}{2x} \times x^2 < (2-x) \times x^2$

$\Rightarrow (x-5)x < 2(2-x)x^2$

$\Rightarrow (x-5)x - 2(2-x)x^2 < 0$

$\Rightarrow \big((x-5) - 2(2-x)x\big)x < 0$

$\Rightarrow (2x^2 - 3x - 5)x < 0$

$\Rightarrow (2x-5)(x+1)x < 0$

Critical values at $x = -1$, $x = 0$ and $x = \frac{5}{2}$

$x < -1$	$-1 < x < 0$	$0 < x < \frac{5}{2}$	$x > \frac{5}{2}$
$-$	$+$	$-$	$+$

$x \in (-\infty, -1) \cup \left(0, \frac{5}{2}\right)$

11) e)

$\frac{x}{x-4} \geq 2x+1$

$\Rightarrow \frac{x}{x-4} \times (x-4)^2 \geq (2x+1) \times (x-4)^2$

$\Rightarrow x(x-4) \geq (2x+1)(x-4)^2$

$\Rightarrow x(x-4) - (2x+1)(x-4)^2 \geq 0$

$\Rightarrow \big(x - (2x+1)(x-4)\big)(x-4) \geq 0$

$\Rightarrow (4+8x-2x^2)(x-4) \geq 0$

$\Rightarrow (2+4x-x^2)(x-4) \geq 0$

Critical values at $x = 2-\sqrt{6}$, $x = 4$ and $x = 2+\sqrt{6}$

$x < 2-\sqrt{6}$	$2-\sqrt{6} < x < 4$	$4 < x < 2+\sqrt{6}$	$x > 2+\sqrt{6}$
$+$	$-$	$+$	$-$

$x \in \big(-\infty, 2-\sqrt{6}\big] \cup \big(4, 2+\sqrt{6}\big]$

11) f)

$\frac{3x}{5-2x} \leq 2x-3$

$\Rightarrow \frac{3x}{5-2x} \times (5-2x)^2 \leq (2x-3) \times (5-2x)^2$

$\Rightarrow 3x(5-2x) \leq (2x-3)(5-2x)^2$

$\Rightarrow 3x(5-2x) - (2x-3)(5-2x)^2 \leq 0$

$\Rightarrow \big(3x - (2x-3)(5-2x)\big)(5-2x) \leq 0$

$\Rightarrow (4x^2-13x+15)(5-2x) \leq 0$

For $4x^2-13x+15$, $b^2-4ac = (-13)^2 - 4(4)(15) = -71 < 0$

so $4x^2-13x+15 > 0$ for all real values of x

$\Rightarrow 5-2x \leq 0$

$\Rightarrow -2x \leq -5$

$\Rightarrow x \geq \frac{5}{2}$

However, $x \neq \frac{5}{2}$ so $x \in \left(\frac{5}{2}, \infty\right)$

11) g)

$\frac{x+2}{x+6} > x - 5$

$\Rightarrow \frac{x+2}{x+6} \times (x+6)^2 > (x-5) \times (x+6)^2$

$\Rightarrow (x+2)(x+6) > (x-5)(x+6)^2$

$\Rightarrow (x+2)(x+6) - (x-5)(x+6)^2 > 0$

$\Rightarrow \big((x+2) - (x-5)(x+6)\big)(x+6) > 0$

$\Rightarrow (32 - x^2)(x+6) > 0$

Critical values at $x = -6$, $x = -4\sqrt{2}$ and $x = 4\sqrt{2}$

$x < -6$	$-6 < x < -4\sqrt{2}$	$-4\sqrt{2} < x < 4\sqrt{2}$	$x > 4\sqrt{2}$
$+$	$-$	$+$	$-$

$x \in (-\infty, -6) \cup \left(-4\sqrt{2}, 4\sqrt{2}\right)$

11) h)

$\frac{4-x}{6-5x} \geq 2x - 1$

$\Rightarrow \frac{4-x}{6-5x} \times (6-5x)^2 \geq (2x-1) \times (6-5x)^2$

$\Rightarrow (4-x)(6-5x) \geq (2x-1)(6-5x)^2$

$\Rightarrow (4-x)(6-5x) - (2x-1)(6-5x)^2 \geq 0$

$\Rightarrow \big((4-x) - (2x-1)(6-5x)\big)(6-5x) \geq 0$

$\Rightarrow (10x^2 - 18x + 10)(6-5x) \geq 0$

$\Rightarrow (5x^2 - 9x + 5)(6-5x) \geq 0$

For $5x^2 - 9x + 5$, $b^2 - 4ac = (-9)^2 - 4(5)(5) = -19 < 0$

so $5x^2 - 9x + 5 > 0$ for all real values of x

$\Rightarrow 6 - 5x \geq 0$

$\Rightarrow -5x \geq -6$

$\Rightarrow x \leq \frac{6}{5}$

However, $x \neq \frac{6}{5}$ so $x \in \left(-\infty, \frac{6}{5}\right)$

11) i)

$\frac{3-2x}{2-3x} \leq 6-x$

$\Rightarrow \frac{3-2x}{2-3x} \times (2-3x)^2 \leq (6-x) \times (2-3x)^2$

$\Rightarrow (3-2x)(2-3x) \leq (6-x)(2-3x)^2$

$\Rightarrow (3-2x)(2-3x) - (6-x)(2-3x)^2 \leq 0$

$\Rightarrow \big((3-2x) - (6-x)(2-3x)\big)(2-3x) \leq 0$

$\Rightarrow (-3x^2+18x-9)(2-3x) \leq 0$

$\Rightarrow (-x^2+6x-3)(2-3x) \leq 0$

Critical values at $x = 3-\sqrt{6}$, $x = \frac{2}{3}$ and $x = 3+\sqrt{6}$

$x < 3-\sqrt{6}$	$3-\sqrt{6} < x < \frac{2}{3}$	$\frac{2}{3} < x < 3+\sqrt{6}$	$x > 3+\sqrt{6}$
$-$	$+$	$-$	$+$

$x \in \left(-\infty, 3-\sqrt{6}\right] \cup \left(\frac{2}{3}, 3+\sqrt{6}\right]$

12) a)

$\frac{(x-1)(x-2)}{x} > 0$

Critical values at $x = 0$, $x = 1$ and $x = 2$

$x < 0$	$0 < x < 1$	$1 < x < 2$	$x > 2$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, 0 < x < 1\} \cup \{x: x \in \mathbb{R}, x > 2\}$

12) b)

$\frac{(x+3)(x-5)}{x} < 0$

Critical values at $x = -3$, $x = 0$ and $x = 5$

$x < -3$	$-3 < x < 0$	$0 < x < 5$	$x > 5$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x < -3\} \cup \{x: x \in \mathbb{R}, 0 < x < 5\}$

12) c)

$\frac{(2x-1)(3x+4)}{2x} \leq 0$

Critical values at $x = -\frac{4}{3}$, $x = 0$ and $x = \frac{1}{2}$

$x < -\frac{4}{3}$	$-\frac{4}{3} < x < 0$	$0 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$-$	$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, x \leq -\frac{4}{3}\right\} \cup \left\{x: x \in \mathbb{R}, 0 < x \leq \frac{1}{2}\right\}$

12) d)

$\frac{(3-x)(1+2x)}{4x} \geq 0$

Critical values at $x = -\frac{1}{2}$, $x = 0$ and $x = 3$

$x < -\frac{1}{2}$	$-\frac{1}{2} < x < 0$	$0 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, x \leq -\frac{1}{2}\right\} \cup \{x: x \in \mathbb{R}, 0 < x \leq 3\}$

13) a)

$\frac{x-3}{(x+1)(x+2)} > 0$

Critical values at $x = -2$, $x = -1$ and $x = 3$

$x < -2$	$-2 < x < -1$	$-1 < x < 3$	$x > 3$
$-$	$+$	$-$	$+$

$x \in (-2, -1) \cup (3, \infty)$

13) b)

$\frac{x+4}{(x-2)(x-5)} < 0$

Critical values at $x = -4$, $x = 2$ and $x = 5$

$x < -4$	$-4 < x < 2$	$2 < x < 5$	$x > 5$
$-$	$+$	$-$	$+$

$x \in (-\infty, -4) \cup (2, 5)$

13) c)

$\frac{2x+1}{(x+3)(x-4)} \geq 0$

Critical values at $x = -3$, $x = -\frac{1}{2}$ and $x = 4$

$x < -3$	$-3 < x < -\frac{1}{2}$	$-\frac{1}{2} < x < 4$	$x > 4$
$-$	$+$	$-$	$+$

$x \in \left(-3, -\frac{1}{2}\right] \cup (4, \infty)$

13) d)

$\frac{3-5x}{(2x+1)(x+3)} \leq 0$

Critical values at $x = -3$, $x = -\frac{1}{2}$ and $x = \frac{3}{5}$

$x < -3$	$-3 < x < -\frac{1}{2}$	$-\frac{1}{2} < x < \frac{3}{5}$	$x > \frac{3}{5}$
$+$	$-$	$+$	$-$

$x \in \left(-3, -\frac{1}{2}\right) \cup \left[\frac{3}{5}, \infty\right)$

14) a)

$\frac{(x-4)(x-3)}{x+2} > 0$

Critical values at $x = -2$, $x = 3$ and $x = 4$

$x < -2$	$-2 < x < 3$	$3 < x < 4$	$x > 4$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, -2 < x < 3\} \cup \{x: x \in \mathbb{R}, x > 4\}$

14) b)

$\frac{(x+1)(x+5)}{x-3} < 0$

Critical values at $x = -5$, $x = -1$ and $x = 3$

$x < -5$	$-5 < x < -1$	$-1 < x < 3$	$x > 3$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x < -5\} \cup \{x: x \in \mathbb{R}, -1 < x < 3\}$

14) c)

$\frac{(2x-3)(x+2)}{x-1} \geq 0$

Critical values at $x = -2$, $x = 1$ and $x = \frac{3}{2}$

$x < -2$	$-2 < x < 1$	$1 < x < \frac{3}{2}$	$x > \frac{3}{2}$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, -2 \leq x < 1\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{3}{2}\right\}$

14) d)

$\frac{(4-x)(x-6)}{2x+3} \leq 0$

Critical values at $x = -\frac{3}{2}$, $x = 4$ and $x = 6$

$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 4$	$4 < x < 6$	$x > 6$
$+$	$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, -\frac{3}{2} < x \leq 4\right\} \cup \{x: x \in \mathbb{R}, x \geq 6\}$

15) a)

$\frac{x-3}{x+5} > x + 12$

$\Rightarrow \frac{x-3}{x+5} - (x + 12) > 0$

$\Rightarrow \frac{x-3}{x+5} - \frac{(x+12)(x+5)}{x+5} > 0$

$\Rightarrow \frac{(x-3)-(x+12)(x+5)}{x+5} > 0$

$\Rightarrow \frac{-x^2+16x+63}{x+5} > 0$

$\Rightarrow -\frac{(x+9)(x+7)}{x+5} > 0$

$\Rightarrow \frac{(x+9)(x+7)}{x+5} < 0$

$\therefore k = 7$

15) b)

Critical values at $x = -9$, $x = -7$ and $x = -5$

$x < -9$	$-9 < x < -7$	$-7 < x < -5$	$x > -5$
$-$	$+$	$-$	$+$

$x \in (-\infty, -9) \cup (-7, -5)$

16) a)

$\frac{(3x-1)(x+1)}{x+2} \geq 12$

$\Rightarrow \frac{(3x-1)(x+1)}{x+2} \times (x+2)^2 \geq 12 \times (x+2)^2$

$\Rightarrow (3x-1)(x+1)(x+2) \geq 12(x+2)^2$

$\Rightarrow (3x-1)(x+1)(x+2) - 12(x+2)^2 \geq 0$

$\Rightarrow \big((3x-1)(x+1) - 12(x+2)\big)(x+2) \geq 0$

$\Rightarrow (3x^2 - 10x - 25)(x+2) \geq 0$

$\Rightarrow (3x+5)(x-5)(x+2) \geq 0$

Critical values at $x = -2$, $x = -\frac{5}{3}$ and $x = 5$

$x < -2$	$-2 < x < -\frac{5}{3}$	$-\frac{5}{3} < x < 5$	$x > 5$
$-$	$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, -2 < x \leq -\frac{5}{3}\right\} \cup \{x: x \in \mathbb{R}, x \geq 5\}$

16) b)

$\frac{(8-x)(2x+5)}{3x+1} \leq 4$

$\Rightarrow \frac{(8-x)(2x+5)}{3x+1} \times (3x+1)^2 \leq 4 \times (3x+1)^2$

$\Rightarrow (8-x)(2x+5)(3x+1) \leq 4(3x+1)^2$

$\Rightarrow (8-x)(2x+5)(3x+1) - 4(3x+1)^2 \leq 0$

$\Rightarrow \big((8-x)(2x+5) - 4(3x+1)\big)(3x+1) \leq 0$

$\Rightarrow (-2x^2 - x + 36)(3x+1) \leq 0$

$\Rightarrow (2x+9)(4-x)(3x+1) \leq 0$

Critical values at $x = -\frac{9}{2}$, $x = -\frac{1}{3}$ and $x = 4$

$x < -\frac{9}{2}$	$-\frac{9}{2} < x < -\frac{1}{3}$	$-\frac{1}{3} < x < 4$	$x > 4$
$+$	$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, -\frac{9}{2} \leq x < -\frac{1}{3}\right\} \cup \{x: x \in \mathbb{R}, x \geq 4\}$

16) c)

$\frac{(2-5x)(4-x)}{6-x} < -4$

$\Rightarrow \frac{(2-5x)(4-x)}{6-x} \times (6-x)^2 < -4 \times (6-x)^2$

$\Rightarrow (2-5x)(4-x)(6-x) < -4(6-x)^2$

$\Rightarrow (2-5x)(4-x)(6-x) + 4(6-x)^2 < 0$

$\Rightarrow \big((2-5x)(4-x) + 4(6-x)\big)(6-x) < 0$

$\Rightarrow (5x^2 - 26x + 32)(6-x) < 0$

$\Rightarrow (5x-16)(x-2)(6-x) < 0$

Critical values at $x = 2$, $x = \frac{16}{5}$ and $x = 6$

$x < -2$	$2 < x < \frac{16}{5}$	$\frac{16}{5} < x < 6$	$x > 6$
$+$	$-$	$+$	$-$

$\left\{x: x \in \mathbb{R}, 2 < x < \frac{16}{5}\right\} \cup \{x: x \in \mathbb{R}, x > 6\}$

17) a)

$\frac{x^2-9}{5x-2} \leq 3 - x$

$\Rightarrow \frac{x^2-9}{5x-2} + x - 3 \leq 0$

$\Rightarrow \frac{(x-3)(x+3)}{5x-2} + \frac{(x-3)(5x-2)}{5x-2} \leq 0$

$\Rightarrow \frac{(x-3)(x+3)+(x-3)(5x-2)}{5x-2} \leq 0$

$\Rightarrow \frac{\big((x+3)+(5x-2)\big)(x-3)}{5x-2} \leq 0$

$\Rightarrow \frac{(6x+1)(x-3)}{5x-2} \leq 0$

$\therefore p = 6$ and $q = 1$

17) b)

Critical values at $x = -\frac{1}{6}$, $x = \frac{2}{5}$ and $x = 3$

$x < -\frac{1}{6}$	$-\frac{1}{6} < x < \frac{2}{5}$	$\frac{2}{5} < x < 3$	$x > 3$
$-$	$+$	$-$	$+$

$x \in \left(\infty, -\frac{1}{6}\right] \cup \left(\frac{2}{5}, 3\right]$

18) a)

$\frac{(x-2)(x-6)}{x-8} \geq 0$ so $a = 2$, $b = 6$, $c = 8$

18) b)

$\frac{(x+3)(x-5)}{x-2} \geq 0$ so $a = -3$, $b = 5$, $c = 2$

19) a)

$\frac{(4-x)(x+2)}{x+4} \geq 0$ so $a = 4$, $b = -2$, $c = -4$

19) b)

$\frac{(7-x)(x-5)}{x-6} \geq 0$ so $a = 7$, $b = 5$, $c = 6$

20) a)

$\frac{(x+1)(x+2)}{(x+3)(x+4)} < 0$

Critical values at $x = -4$, $x = -3$, $x = -2$ and $x = -1$

$x < -4$	$-4 < x < -3$	$-3 < x < -2$	$-2 < x < -1$	$x > -1$
$+$	$-$	$+$	$-$	$+$

$x \in (-4, -3) \cup (-2, -1)$

20) b)

$\frac{(x+2)(4-x)}{(x-1)(x+3)} \geq 0$

Critical values at $x = -3$, $x = -2$, $x = 1$ and $x = 4$

$x < -3$	$-3 < x < -2$	$-2 < x < 1$	$1 < x < 4$	$x > 4$
$-$	$+$	$-$	$+$	$-$

$x \in (-3, -2] \cup (1, 4]$

20) c)

$\frac{(x-3)(5-x)}{(2-x)(6-x)} \leq 0$

Critical values at $x = 2$, $x = 3$, $x = 5$ and $x = 6$

$x < 2$	$2 < x < 3$	$3 < x < 5$	$5 < x < 6$	$x > 6$
$-$	$+$	$-$	$+$	$-$

$x \in (-\infty, 2) \cup [3, 5] \cup (6, \infty)$

21)

$e^x + 2 > 2$ for all real values of x

$\frac{x+5}{2x-2} > 2$

$\Rightarrow \frac{x+5}{2(x-1)} > 2$

$\Rightarrow \frac{x+5}{x-1} > 4$

$\Rightarrow \frac{x+5}{x-1} \times (x-1)^2 > 4 \times (x-1)^2$

$\Rightarrow (x+5)(x-1) > 4(x-1)^2$

$\Rightarrow (x+5)(x-1) - 4(x-1)^2 > 0$

$\Rightarrow \big((x+5) - 4(x-1)\big)(x-1) > 0$

$\Rightarrow (9-3x)(x-1) > 0$

$\Rightarrow (3-x)(x-1) > 0$

Critical values at $x = 1$ and $x = 3$

$x < 1$	$1 < x < 3$	$x > 3$
$-$	$+$	$-$

$\{x: x \in \mathbb{R}, 1 < x < 3\}$

22) a)

$\frac{2x}{x-k} > x + k$

$\Rightarrow \frac{2x}{x-k} \times (x-k)^2 > (x+k) \times (x-k)^2$

$\Rightarrow 2x(x-k) > (x+k)(x-k)^2$

$\Rightarrow 2x(x-k) - (x+k)(x-k)^2 > 0$

$\Rightarrow \big(2x - (x+k)(x-k)\big)(x-k) > 0$

$\Rightarrow (-x^2 + 2x + k^2)(x-k) > 0$

Critical values at $x = 1 - \sqrt{k^2+1}$, $x = k$ and $x = 1 + \sqrt{k^2+1}$

$x < 1 - \sqrt{k^2+1}$	$1 - \sqrt{k^2+1} < x < k$	$k < x < 1 + \sqrt{k^2+1}$	$x > 1 + \sqrt{k^2+1}$
$+$	$-$	$+$	$-$

$\{x : x \in \mathbb{R}, x < 1 - \sqrt{k^2+1}\} \cup \{x : x \in \mathbb{R}, k < x < 1 + \sqrt{k^2+1}\}$

22) b)

$\frac{x-4}{k-x} < 4 - x$

$\Rightarrow \frac{x-4}{k-x} \times (k-x)^2 < (4-x) \times (k-x)^2$

$\Rightarrow (x-4)(k-x) < (4-x)(k-x)^2$

$\Rightarrow (x-4)(k-x) - (4-x)(k-x)^2 < 0$

$\Rightarrow (x-4)(k-x) + (x-4)(k-x)^2 < 0$

$\Rightarrow \big(1 + (k-x)\big)(x-4)(k-x) < 0$

$\Rightarrow (k+1-x)(x-4)(k-x) < 0$

Critical values at $x = 4$, $x = k$ and $x = k + 1$

$x < 4$	$4 < x < k$	$k < x < k+1$	$x > k+1$
$-$	$+$	$-$	$+$

$\{x : x \in \mathbb{R}, x < 4\} \cup \{x : x \in \mathbb{R}, k < x < k + 1\}$

22) c)

$\frac{x^2-k^2}{x+3} \geq k - x$

$\Rightarrow \frac{x^2-k^2}{x+3} \times (x+3)^2 \geq (k-x) \times (x+3)^2$

$\Rightarrow (x^2-k^2)(x+3) \geq (k-x)(x+3)^2$

$\Rightarrow (x^2-k^2)(x+3) - (k-x)(x+3)^2 \geq 0$

$\Rightarrow (x-k)(x+k)(x+3) + (x-k)(x+3)^2 \geq 0$

$\Rightarrow \big((x+k)+(x+3)\big)(x-k)(x+3) \geq 0$

$\Rightarrow (2x+k+3)(x-k)(x+3) \geq 0$

Critical values at $x = -\frac{k+3}{2}$, $x = -3$ and $x = k$

$x < -\frac{k+3}{2}$	$-\frac{k+3}{2} < x < -3$	$-3 < x < k$	$x > k$
$-$	$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, -\frac{k+3}{2} \leq x < -3\right\} \cup \{x: x \in \mathbb{R}, x \geq k\}$